

Workshop Preparation Guide

A Practical Introduction to 3DP-Jmol and 3D Printing

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Introduction

This workshop is designed to provide participants with both the theoretical background and practical skills required to generate and 3D print molecular models using and [3DP-Jmol](#) and its web implementation - [EDUMOL3D](#) tool. Through guided exercises, participants will explore how 3D-printed macromolecular models can enhance teaching, communication, and research in the molecular life sciences.

To ensure that all attendees can fully benefit from the hands-on component of the session, a software tool needs to be installed in advance, and a few files must be downloaded. The following document provides step-by-step instructions for:

- Required computer hardware;
- setting up the required software environment;
- preparing necessary files to use during the workshop.

Please make sure that you complete all setup steps before arriving at the session for a smooth experience.

1. Computer Hardware Requirements

Attendees must bring a personal **laptop computer** running Windows, macOS, or Linux. Most modern laptops will work perfectly well, so there is no need to worry about performance requirements. A **mouse** (wired or wireless) is **strongly recommended**, as precise model manipulation is difficult using a touchpad alone. And most importantly — don't forget to bring your **laptop charger**!

2. Required Software

During the workshop, we will use one single software tool that needs installing:

Bambu Studio

Purpose: **Bambu Studio** is a slicer program used to prepare 3D printable models for the printer. It converts .STL files into printer instructions (.gcode).

Download link: <https://bambulab.com/en/download/studio>

Installation instructions:

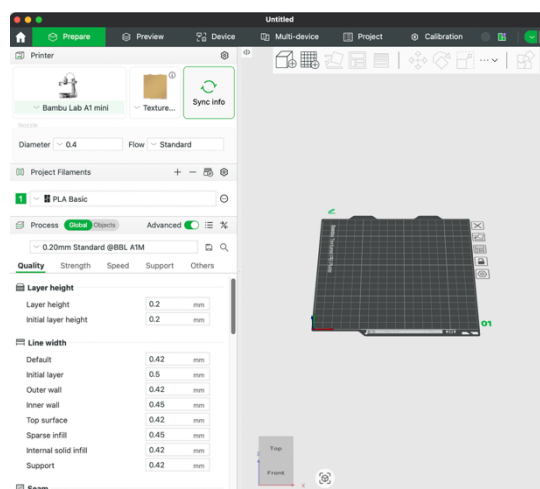
- a. Visit the link above and select the installer for your operating system.
- b. Download and run the installer following the on-screen instructions.

c. When first launched, run the **Configuration Wizard** and accept all default options, **except for the printer selection screen**:

- Select “Bambu Lab A1 Mini” (used for demonstration).

d. Once installation is complete, close the program — it will be configured later during the workshop.

Check if everything works: Launch the **Bambu Studio** from the Start Menu/Dock/Launcher. If prompted to register a MakerWorld account, simply cancel — registration is **not required** for this workshop. In the end You should see an window as shown on the right. This means it is ready for use during the workshop.



3. Workshop Files and Folder Setup

Purpose: To ensure a smooth start during the practical part of the workshop, you will need a configuration file for Bambu Studio (**.bbscfg**). The file is provided as a compressed archive (**.zip**) available for download from the link below.

4.1. Download the Workshop Package

Download link:

[https://github.com/mariusmihasan/3DP-](https://github.com/mariusmihasan/3DP-Jmol/blob/devel/workshops/2026_Suceava/2026_Suceava_files.zip)

[Jmol/blob/devel/workshops/2026_Suceava/2026_Suceava_files.zip](https://github.com/mariusmihasan/3DP-Jmol/blob/devel/workshops/2026_Suceava/2026_Suceava_files.zip)

Click on the link above to download the **2026_Suceava_files.zip** archive to your computer. Save the file to a convenient location (for example, your Desktop or Documents folder).

4.2. Unpack and Store the Files

Once the download is complete, unzip the archive (usually by double-clicking it or right-clicking on it → Extract All).

You should now see **FIVE** (5) files:

1. **Bambu Lab A1 mini 0.4 nozzle.bbscfg** – configuration bundle to be imported later into Bambu Studio;
2. **B_DNA_surface (repaired).stl**
3. **bdna_color_A1mini.3mf**
4. **bdna_color.obj**
5. **bdna_color.mtl**

Files 2 to 5 are a bunch of back-up files in case stuff does not work as expected 😊.

All materials, including this guide and the slides used for the presentation and workshop are available at:

https://github.com/mariusmihasan/3DP-Jmol/tree/devel/workshops/2026_Suceava

5. Feedback and Follow-Up – please scroll down.

Your feedback is extremely valuable in improving future workshops and ensuring that the 3DP-Jmol resources continue to evolve to best serve the educational community.

Please take 2–3 minutes after the session to complete a short, anonymous feedback form:

 Feedback Form: <https://forms.gle/zUAG9ChebcJdkedP8>

 Estimated time: 2–3 minutes

You may also access the form by scanning the QR code:



Your responses will help us refine future workshops, improve the clarity of installation materials, and design new teaching activities using 3D-printed molecular models.

Thank you for your time and participation!

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